

VisitorQR

Visitor Management System

REST API Documentation

Version 2.2 | FastAPI + SQLAlchemy + SQLite
Base URL: <http://localhost:8000/api>

Authentication: JWT Bearer Token

Interactive Docs: <http://localhost:8000/docs>
Roles: guard | admin | super_admin

Field	Details
Document Type	REST API Documentation
System Name	QR-Based Visitor Management System
Version	v2.0.0
Prepared By	Amit Kushwaha
Department	MP Online Limited
Date	27 April 2026
Project Type	Internship Project

1. Overview

The VisitorQR API is a RESTful backend for a QR-based Visitor Management System. It enables visitor self-registration, QR code generation and delivery, smart check-in/check-out via QR scan, and full administrative control over users, branches, and audit logs.

1.1 Base URL & Format

```
Base URL : http://localhost:8000/api
Format   : All requests and responses use JSON (application/json)
Auth     : Bearer JWT token in Authorization header
Docs     : http://localhost:8000/docs (Swagger UI)
```

1.2 User Roles

All protected endpoints enforce role-based access control (RBAC). Three roles exist:

Role	Access Level	Capabilities
guard	Limited	Scan QR codes to check visitors in and out. View visit list.
admin	Standard	All guard permissions + Dashboard, Reports, Visitor management, Branch listing.
super_admin	Full	All admin permissions + User management, Branch creation, Audit log access.

1.3 Authentication

Obtain a token by calling POST /auth/login with form data (username + password). Include the token in all protected requests:

```
Authorization: Bearer <your_jwt_token>
```

Tokens expire after 24 hours. A 401 response means the token is missing, invalid, or expired.

1.4 HTTP Status Codes

Code	Meaning
200 OK	Request succeeded. Response body contains the result.
201 Created	Resource created successfully.

Code	Meaning
204 No Content	Success with no response body (e.g. delete).
400 Bad Request	Invalid input data or validation error.
401 Unauthorized	Missing or invalid JWT token.
403 Forbidden	Authenticated but insufficient role.
404 Not Found	Requested resource does not exist.
422 Unprocessable	Request body failed schema validation (Pydantic).
500 Server Error	Unexpected server-side error.

2. Endpoint Summary

All endpoints are prefixed with /api. The table below provides a quick reference of every available route.

Method	Endpoint	Description	Auth Required
POST	/auth/login	Login and receive a JWT token	Public
GET	/auth/me	Get the logged-in user profile	Any role
PUT	/auth/profile	Update own name / phone / department	Any role
POST	/auth/change-password	Change own password	Any role
POST	/visitors/register	Register a visitor and generate QR pass	Public
GET	/visitors	List all visitors	guard+
GET	/visitors/{visitor_id}	Get a single visitor by ID	guard+
POST	/visits/scan/{qr_token}	Scan QR to check-in or check-out	guard+
GET	/visits	List all visits (filter by branch/status)	guard+
GET	/visits/{visit_id}	Get a single visit by ID	guard+
GET	/admin/users	List all system users	super_admin
POST	/admin/users	Create a new system user	super_admin
DELETE	/admin/users/{user_id}	Deactivate a user account	super_admin
GET	/admin/branches	List all active office branches	Public
POST	/admin/branches	Create a new branch	super_admin
GET	/admin/dashboard	Get real-time dashboard statistics	admin+
GET	/admin/report	Get visit analytics report	admin+
GET	/admin/audit-logs	List all audit log entries	super_admin
GET	/admin/audit-logs/search	Search audit logs with filters	super_admin

3. Authentication Endpoints

Auth endpoints handle login, token refresh, profile management, and password changes.

3.1 POST /auth/login

Description

Authenticate with email and password. Returns a JWT Bearer token valid for 24 hours. Uses OAuth2 form data format (not JSON).

Request (form-urlencoded)

Field	Type	Required	Description
username	string (email)	Yes	The user's email address (used as username)
password	string	Yes	The user's plaintext password

Response — 200 OK

Field	Type	Required	Description
access_token	string	—	JWT Bearer token. Include in Authorization header.
token_type	string	—	Always 'bearer'
user_id	integer	—	Unique user ID
user_name	string	—	Full name of the logged-in user
user_role	string	—	Role: guard admin super_admin
branch_id	integer null	—	Branch the user belongs to (null if unassigned)

Example Request

```
POST /api/auth/login
Content-Type: application/x-www-form-urlencoded

username=admin@company.com&password=secret123
```

Example Response

```
{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer", "user_id": 1,
  "user_name": "System Admin", "user_role": "super_admin", "branch_id": null }
```

3.2 GET /auth/me

Description

Returns the full profile of the currently authenticated user. No request body needed.

Response — 200 OK

Field	Type	Required	Description
<code>id</code>	<code>integer</code>	—	User's unique ID
<code>full_name</code>	<code>string</code>	—	Full name
<code>email</code>	<code>string</code>	—	Email address (used to login)
<code>phone</code>	<code>string null</code>	—	Optional phone number
<code>department</code>	<code>string null</code>	—	Optional department name
<code>role</code>	<code>string</code>	—	guard admin super_admin
<code>branch_id</code>	<code>integer null</code>	—	Branch ID or null
<code>is_active</code>	<code>boolean</code>	—	Whether the account is active
<code>created_at</code>	<code>datetime</code>	—	Account creation timestamp (UTC ISO 8601)

3.3 PUT /auth/profile

Description

Update the logged-in user's own display name, phone number, or department. Role and branch can only be changed by a super_admin.

Request Body (JSON)

Field	Type	Required	Description
<code>full_name</code>	<code>string</code>	No	New display name (all fields optional)
<code>phone</code>	<code>string</code>	No	New phone number
<code>department</code>	<code>string</code>	No	New department name

3.4 POST /auth/change-password

Description

Change the authenticated user's own password. The current password is required for verification.

Request Body (JSON)

Field	Type	Required	Description
<code>current_password</code>	<code>string</code>	Yes	The user's existing password for verification
<code>new_password</code>	<code>string</code>	Yes	The desired new password (min 8 characters recommended)

4. Visitor Registration Endpoints

These endpoints handle the core visitor lifecycle — registration and QR pass generation.

4.1 POST/visitors/register

Description

PUBLIC endpoint — no authentication needed. Registers a new visitor, creates a visit record, generates a unique QR code, and optionally emails the QR pass to the visitor. Admins and super_admins can additionally set custom QR validity duration.

Request Body (JSON)

Field	Type	Required	Description
<code>full_name</code>	<code>string</code>	Yes	Visitor's full name
<code>phone</code>	<code>string</code>	Yes	Contact phone number
<code>email</code>	<code>string</code>	No	Email address — QR code will be emailed here if SMTP is configured
<code>company_name</code>	<code>string</code>	No	Visitor's company or organization
<code>host_name</code>	<code>string</code>	No	Name of the employee being visited
<code>purpose</code>	<code>string</code>	No	Visit purpose: meeting delivery interview maintenance other (default: meeting)
<code>branch_id</code>	<code>integer</code>	Yes	ID of the office branch being visited
<code>notes</code>	<code>string</code>	No	Additional notes or comments
<code>photo_data</code>	<code>string</code>	No	Base64 data URI of visitor photo (e.g. 'data:image/jpeg;base64,...')
<code>qr_validity_type</code>	<code>string</code>	No	Admin only: one_time hourly daily weekly monthly (default: uses .env QR_EXPIRY_HOURS)
<code>qr_hours</code>	<code>integer</code>	No	Admin only: number of hours (1–8760), used only when qr_validity_type = hourly

Response — 201 Created

Field	Type	Required	Description
<code>id</code>	<code>integer</code>	—	Visit record ID
<code>visitor_id</code>	<code>integer</code>	—	Visitor record ID
<code>branch_id</code>	<code>integer</code>	—	Branch where visit is registered
<code>host_name</code>	<code>string null</code>	—	Person being visited

Field	Type	Required	Description
<code>purpose</code>	<code>string</code>	—	Visit purpose value
<code>qr_token</code>	<code>string</code>	—	Unique token embedded in the QR code
<code>qr_image</code>	<code>string</code>	—	Base64 data URI of the QR code PNG — use as <code></code> in frontend
<code>qr_expires</code>	<code>datetime</code>	—	QR code expiry timestamp (UTC ISO 8601)
<code>qr_expiry_hours</code>	<code>integer null</code>	—	Hours until QR expires
<code>qr_validity_type</code>	<code>string</code>	—	<code>hourly</code> <code>daily</code> <code>weekly</code> <code>monthly</code> <code>one_time</code>
<code>status</code>	<code>string</code>	—	<code>registered</code> <code>checked_in</code> <code>checked_out</code> <code>expired</code>
<code>registered_at</code>	<code>datetime</code>	—	Registration timestamp (UTC ISO 8601)
<code>visitor</code>	<code>object</code>	—	Nested visitor object: <code>id</code> , <code>full_name</code> , <code>phone</code> , <code>email</code> , <code>company_name</code> , <code>photo_path</code>
<code>email_sent</code>	<code>boolean</code>	—	True if QR email was sent successfully
<code>email_address</code>	<code>string</code>	—	Email address the QR was sent to (empty string if not sent)

QR Validity Options (Admin Only)

<code>qr_validity_type</code>	Duration	Behaviour
<code>(not set)</code>	From <code>.env</code>	Uses <code>QR_EXPIRY_HOURS</code> value set in backend <code>.env</code> file
<code>one_time</code>	Variable	QR becomes invalid immediately after the visitor checks out
<code>hourly</code>	1–8760 hrs	Custom hours specified in <code>qr_hours</code> field
<code>daily</code>	24 hours	QR is valid for exactly 24 hours from registration
<code>weekly</code>	7 days	QR is valid for 168 hours (7 days) from registration
<code>monthly</code>	30 days	QR is valid for 720 hours (30 days) from registration

4.2 GET /visitors

Description

Returns a paginated list of all registered visitors. Requires guard role or higher.

Query Parameters

Field	Type	Required	Description
<code>skip</code>	<code>integer</code>	No	Number of records to skip (default: 0)

Field	Type	Required	Description
<code>limit</code>	<code>integer</code>	No	Maximum records to return (default: 50, max: 200)

4.3 GET /visitors/{visitor_id}

Description

Returns the profile of a single visitor by their ID. Returns 404 if not found.

Path Parameter

Field	Type	Required	Description
<code>visitor_id</code>	<code>integer</code>	Yes	The unique ID of the visitor to retrieve

5. Visit & Scan Endpoints

These endpoints handle the physical check-in and check-out process at the office entrance via QR code scanning.

5.1 POST /visits/scan/{qr_token}

Description

Smart scan endpoint. The guard scans the visitor's QR code. The system automatically determines the action: - If the visitor has not checked in yet → performs Check-In - If the visitor is already checked in → performs Check-Out The scanning guard's user ID is logged for audit purposes.

Path Parameter

Field	Type	Required	Description
<code>qr_token</code>	<code>string</code>	Yes	The unique token extracted from the visitor's QR code

Response — 200 OK

Field	Type	Required	Description
<code>message</code>	<code>string</code>	—	Human-readable result: 'Checked In' or 'Checked Out'
<code>visit_id</code>	<code>integer</code>	—	The visit record ID
<code>visitor_name</code>	<code>string</code>	—	Full name of the visitor
<code>visitor_phone</code>	<code>string</code>	—	Visitor's phone number
<code>host_name</code>	<code>string</code>	—	Name of the host being visited
<code>purpose</code>	<code>string</code>	—	Visit purpose
<code>action</code>	<code>string</code>	—	<code>check_in</code> or <code>check_out</code>
<code>timestamp</code>	<code>datetime</code>	—	When the scan occurred (UTC ISO 8601)

Error Responses

Field	Type	Required	Description
<code>404 Not Found</code>	—	—	No visit found with the provided QR token
<code>409 Conflict</code>	—	—	QR code has expired or visitor is already checked out
<code>401 Unauthorized</code>	—	—	Guard is not logged in
<code>403 Forbidden</code>	—	—	Authenticated user does not have scan permission

5.2 GET /visits

Description

Returns a list of visit records. Supports filtering by branch and status.

Query Parameters

Field	Type	Required	Description
<code>branch_id</code>	<code>integer</code>	No	Filter visits by branch ID
<code>status</code>	<code>string</code>	No	Filter by status: registered checked_in checked_out expired
<code>skip</code>	<code>integer</code>	No	Pagination offset (default: 0)
<code>limit</code>	<code>integer</code>	No	Max results (default: 50, max: 200)

5.3 GET /visits/{visit_id}

Description

Returns full details of a single visit record including nested visitor information.

Path Parameter

Field	Type	Required	Description
<code>visit_id</code>	<code>integer</code>	Yes	The unique ID of the visit record

6. Admin Endpoints

Admin endpoints are for user management, branch management, dashboard statistics, and reporting. All require admin or super_admin role unless noted.

6.1 GET /admin/users

Description

Returns a paginated list of all registered system users (guards, admins, super_admins). Requires super_admin.

Query Parameters

Field	Type	Required	Description
skip	integer	No	Pagination offset (default: 0)
limit	integer	No	Max results (default: 100, max: 500)

6.2 POST /admin/users

Description

Creates a new system user account. The password is hashed before storage. Requires super_admin.

Request Body (JSON)

Field	Type	Required	Description
full_name	string	Yes	Full name of the new user
email	string	Yes	Email address (used as login username, must be unique)
password	string	Yes	Initial password (will be bcrypt hashed)
phone	string	No	Optional phone number
department	string	No	Optional department name
role	string	No	guard admin super_admin (default: guard)
branch_id	integer	No	Branch the user will be assigned to

6.3 DELETE /admin/users/{user_id}

Description

Deactivates a user account (soft delete — sets `is_active=false`). The user can no longer log in but their records are preserved for audit purposes. Requires `super_admin`.

Path Parameter

Field	Type	Required	Description
<code>user_id</code>	<code>integer</code>	Yes	The ID of the user to deactivate

6.4 GET /admin/branches

Description

Returns a list of all active office branches. This endpoint is PUBLIC — no authentication required. Used by the visitor registration form to populate the branch dropdown.

Response Fields

Field	Type	Required	Description
<code>id</code>	<code>integer</code>	—	Unique branch ID
<code>name</code>	<code>string</code>	—	Branch name (e.g. 'Mumbai HQ')
<code>city</code>	<code>string null</code>	—	City where the branch is located
<code>address</code>	<code>string null</code>	—	Full address
<code>is_active</code>	<code>boolean</code>	—	Whether the branch is currently active
<code>created_at</code>	<code>datetime</code>	—	When the branch was created (UTC ISO 8601)

6.5 POST /admin/branches

Description

Creates a new office branch. Requires `super_admin`.

Request Body (JSON)

Field	Type	Required	Description
<code>name</code>	<code>string</code>	Yes	Branch name (must be unique and descriptive)
<code>city</code>	<code>string</code>	No	City of the branch
<code>address</code>	<code>string</code>	No	Full postal address

6.6 GET /admin/dashboard

Description

Returns real-time statistics for the office dashboard. Requires admin or super_admin.

Query Parameters

Field	Type	Required	Description
<code>branch_id</code>	<code>integer</code>	No	Filter dashboard stats by a specific branch (null = all branches)

Response Fields

Field	Type	Required	Description
<code>currently_inside</code>	<code>integer</code>	—	Number of visitors currently checked in
<code>total_today</code>	<code>integer</code>	—	Total visits registered today
<code>total_this_week</code>	<code>integer</code>	—	Total visits registered this calendar week
<code>avg_duration_mins</code>	<code>float</code>	—	Average visit duration in minutes
<code>purpose_breakdown</code>	<code>object</code>	—	Count of visits per purpose type
<code>currently_inside_list</code>	<code>array</code>	—	List of visitor details for those currently inside
<code>recent_activity</code>	<code>array</code>	—	Last 20 visit events with timestamps
<code>generated_at</code>	<code>string</code>	—	Timestamp when this data was generated

6.7 GET /admin/report

Description

Returns historical visit analytics for a configurable time period. Requires admin or super_admin.

Query Parameters

Field	Type	Required	Description
<code>branch_id</code>	<code>integer</code>	No	Filter by branch ID (null = all branches)
<code>days</code>	<code>integer</code>	No	Report period in days (min: 7, max: 365, default: 30)

Response Fields

Field	Type	Required	Description
<code>period_days</code>	<code>integer</code>	—	The number of days the report covers
<code>daily_counts</code>	<code>array</code>	—	List of {date, count} objects for each day in the period
<code>top_hosts</code>	<code>array</code>	—	Top 10 most visited hosts with visit counts
<code>total_visits</code>	<code>integer</code>	—	Total visits in the period

7. Audit Log Endpoints

Audit logs record every significant action taken in the system — logins, registrations, check-ins, user changes, etc. Access requires `super_admin` role.

7.1 GET /admin/audit-logs

Description

Returns the most recent audit log entries in reverse chronological order. Requires `super_admin`.

Query Parameters

Field	Type	Required	Description
<code>skip</code>	<code>integer</code>	No	Pagination offset (default: 0)
<code>limit</code>	<code>integer</code>	No	Max results (default: 100, max: 500)

Response — Array of log objects

Field	Type	Required	Description
<code>id</code>	<code>integer</code>	—	Unique log entry ID
<code>user_id</code>	<code>integer null</code>	—	ID of the user who performed the action (null for system actions)
<code>action</code>	<code>string</code>	—	Action name: <code>user_login</code> <code>visitor_registered</code> <code>visitor_checkin</code> <code>visitor_checkout</code> <code>user_created</code> <code>user_deactivated</code> <code>profile_updated</code>
<code>entity</code>	<code>string null</code>	—	Type of record affected: <code>visit</code> <code>visitor</code> <code>user</code> <code>branch</code>
<code>entity_id</code>	<code>integer null</code>	—	ID of the affected record
<code>details</code>	<code>string null</code>	—	Additional context as JSON string or plain text
<code>ip_address</code>	<code>string null</code>	—	IP address of the client that performed the action
<code>created_at</code>	<code>datetime</code>	—	When the action occurred (UTC ISO 8601)

7.2 GET /admin/audit-logs/search

Description

Advanced filtered search across audit logs. All filter parameters are optional and can be combined freely. Requires `super_admin`.

Query Parameters

Field	Type	Required	Description
<code>user_id</code>	<code>integer</code>	No	Filter logs by a specific user's ID
<code>action</code>	<code>string</code>	No	Partial match on action name (e.g. 'login' matches 'user_login')
<code>entity</code>	<code>string</code>	No	Filter by entity type: visit visitor user branch
<code>entity_id</code>	<code>integer</code>	No	Filter by a specific entity's ID
<code>date</code>	<code>string</code>	No	Exact date filter (YYYY-MM-DD format) — returns logs for that day only
<code>date_from</code>	<code>string</code>	No	Start of date range (YYYY-MM-DD). Combine with <code>date_to</code> for a range.
<code>date_to</code>	<code>string</code>	No	End of date range (YYYY-MM-DD, inclusive up to 23:59:59)
<code>last_n_days</code>	<code>integer</code>	No	Shortcut: return logs from the last N days (e.g. 7 for last week)
<code>time_from</code>	<code>string</code>	No	Filter by time of day from (HH:MM format, applies to <code>created_at</code>)
<code>time_to</code>	<code>string</code>	No	Filter by time of day to (HH:MM format)
<code>ip_address</code>	<code>string</code>	No	Partial match on IP address (e.g. '192.168' matches any 192.168.x.x)
<code>sort_order</code>	<code>string</code>	No	asc for oldest first, desc for newest first (default: desc)
<code>skip</code>	<code>integer</code>	No	Pagination offset (default: 0)
<code>limit</code>	<code>integer</code>	No	Max results (default: 50, max: 500)

Response

Field	Type	Required	Description
<code>total</code>	<code>integer</code>	—	Total number of matching records (before pagination)
<code>skip</code>	<code>integer</code>	—	Offset that was applied
<code>limit</code>	<code>integer</code>	—	Limit that was applied
<code>results</code>	<code>array</code>	—	Array of matching audit log objects (same fields as GET /admin/audit-logs)

Example Search Requests

```
GET /api/admin/audit-logs/search?action=login&last_n_days=7
GET /api/admin/audit-logs/search?date=2026-04-24&entity=visit
```

```
GET /api/admin/audit-logs/search?user_id=3&date_from=2026-04-01&date_to=2026-04-30
```

```
GET /api/admin/audit-logs/search?time_from=09:00&time_to=17:00&sort_order=asc
```

8. Data Models

The following tables describe the database schema used by the VMS backend.

8.1 users

Column	Type	Description
<code>id</code>	<code>INTEGER PK</code>	Auto-increment primary key
<code>full_name</code>	<code>VARCHAR(100)</code>	User's full name
<code>email</code>	<code>VARCHAR(150)</code>	Unique email address (login username)
<code>hashed_password</code>	<code>VARCHAR(255)</code>	bcrypt-hashed password
<code>phone</code>	<code>VARCHAR(20)</code>	Optional phone number
<code>department</code>	<code>VARCHAR(80)</code>	Optional department
<code>role</code>	<code>ENUM</code>	guard admin super_admin
<code>branch_id</code>	<code>INTEGER FK</code>	References branches.id (nullable)
<code>is_active</code>	<code>BOOLEAN</code>	False = deactivated, cannot login
<code>created_at</code>	<code>DATETIME</code>	Account creation timestamp (UTC)
<code>last_login</code>	<code>DATETIME</code>	Last successful login timestamp (UTC, nullable)

8.2 visitors

Column	Type	Description
<code>id</code>	<code>INTEGER PK</code>	Auto-increment primary key
<code>full_name</code>	<code>VARCHAR(100)</code>	Visitor's full name
<code>phone</code>	<code>VARCHAR(20)</code>	Phone number
<code>email</code>	<code>VARCHAR(150)</code>	Optional email address
<code>company_name</code>	<code>VARCHAR(120)</code>	Optional company or organization
<code>photo_path</code>	<code>VARCHAR(255)</code>	Relative path to visitor photo in uploads/photos/
<code>created_at</code>	<code>DATETIME</code>	First registration timestamp (UTC)

8.3 visits

Column	Type	Description
<code>id</code>	<code>INTEGER PK</code>	Auto-increment primary key

Column	Type	Description
<code>visitor_id</code>	INTEGER FK	References visitors.id
<code>branch_id</code>	INTEGER FK	References branches.id
<code>host_name</code>	VARCHAR(100)	Name of employee being visited
<code>purpose</code>	ENUM	meeting delivery interview maintenance other
<code>notes</code>	TEXT	Optional additional notes
<code>qr_token</code>	VARCHAR(64)	Unique token embedded in the QR code
<code>qr_image</code>	TEXT	Base64 PNG data URI of the generated QR code
<code>qr_expires</code>	DATETIME	QR expiry timestamp (UTC)
<code>qr_expiry_hours</code>	INTEGER	Hours until expiry (stored for audit reference)
<code>status</code>	ENUM	registered checked_in checked_out expired
<code>scanned_by</code>	INTEGER FK	User ID of guard who scanned (nullable)
<code>checked_in_at</code>	DATETIME	Check-in timestamp (UTC, nullable)
<code>checked_out_at</code>	DATETIME	Check-out timestamp (UTC, nullable)
<code>duration_mins</code>	INTEGER	Duration of visit in minutes (calculated on checkout)
<code>registered_at</code>	DATETIME	Visit registration timestamp (UTC)

9. Error Handling

All errors follow a consistent JSON format. The detail field contains a human-readable description of the error.

Standard Error Response Format

```
{ "detail": "Incorrect email or password." }
```

Validation Error Format (422)

```
{ "detail": [{ "loc": ["body", "email"], "msg": "value is not a valid email address",  
"type": "value_error" }] }
```

Common Error Scenarios

Status	detail message	When it occurs
401	Incorrect email or password.	Login with wrong credentials
401	Could not validate credentials.	JWT token is expired or malformed
403	Access denied. Required role...	User's role is insufficient
404	Visit not found.	QR token or visit ID does not exist
404	Visitor not found.	Visitor ID does not exist
409	QR code has expired.	Visitor tried to use an expired QR
409	Visitor already checked out.	QR scanned after visitor already left
422	(Pydantic validation array)	Missing required field or wrong data type
500	(Internal server error)	Unexpected server-side exception

10. Environment Configuration

The backend is configured via a `.env` file in the `backend/` directory. Copy `.env.example` and fill in the values.

Variable	Default	Description
<code>DATABASE_URL</code>	<code>sqlite:///./vms.db</code>	SQLAlchemy database connection string
<code>SECRET_KEY</code>	<code>(required)</code>	Random secret for JWT signing (min 32 chars recommended)
<code>ACCESS_TOKEN_EXPIRE_MINUTES</code>	<code>1440</code>	JWT token lifetime in minutes (1440 = 24 hours)
<code>QR_EXPIRY_HOURS</code>	<code>24</code>	Default QR code validity in hours
<code>SMTP_ENABLED</code>	<code>false</code>	Set to true to enable email delivery of QR codes
<code>SMTP_HOST</code>	<code>smtp.gmail.com</code>	SMTP server hostname
<code>SMTP_PORT</code>	<code>587</code>	SMTP port (587 for TLS, 465 for SSL)
<code>SMTP_USER</code>	<code>(your email)</code>	SMTP authentication username / email address
<code>SMTP_PASSWORD</code>	<code>(app password)</code>	SMTP authentication password (use App Password for Gmail)
<code>SMTP_FROM</code>	<code>(your email)</code>	From address shown in sent emails
<code>APP_NAME</code>	<code>VisitorQR</code>	Application name shown in emails and API title
<code>APP_VERSION</code>	<code>2.2</code>	API version string

11. Quick Start Guide

Step 1 — Start the Backend

```
cd vms_upgraded/backend
python -m venv venv
venv\Scripts\activate      # Windows
pip install -r requirements.txt
uvicorn main:app --reload
# API now running at http://localhost:8000
```

Step 2 — Login

```
POST http://localhost:8000/api/auth/login
Content-Type: application/x-www-form-urlencoded
username=admin@company.com&password=your_password
```

Step 3 — Register a Visitor

```
POST http://localhost:8000/api/visitors/register
Content-Type: application/json

{ "full_name": "Ramesh Kumar",
  "phone": "9876543210",
  "email": "ramesh@example.com",
  "purpose": "meeting",
  "branch_id": 1,
  "host_name": "Priya Sharma" }
```

Step 4 — Scan QR at Entry

```
POST http://localhost:8000/api/visits/scan/{qr_token}
Authorization: Bearer <guard_jwt_token>
```

Step 5 — View Dashboard

```
GET http://localhost:8000/api/admin/dashboard
Authorization: Bearer <admin_jwt_token>
```

Interactive API Explorer

Open <http://localhost:8000/docs> in your browser for the full Swagger UI — you can test every endpoint directly from the browser without Postman.

End of VisitorQR API Documentation — Version 2.2

For support, refer to <http://localhost:8000/docs> or contact the system administrator.